

Space, Time, and Memory

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Time by itself does not exist . . . It must not be claimed that anyone can sense time apart from the movement of things (Lucretius, Book 1).

Introduction

Nothing is more intuitive, yet more complex, than the concepts of space and time. We can live without sight, sound, smell, touch, or even the ability to move, but everything we do feels to occur in space and time. These concepts are built in our language and thinking. For most cultures, space and time are used to explain the contrast between us as individuals against the vastness and complexities of the universe. Time and space are often used interchangeably ('my lab is 5 minutes from here'; Boroditsky 2018) and 'time' is among the most frequently used words in all cultures. Time is intimately related to movement and geometry. Scientific thinking has radically transformed these dimensionless concepts with the introduction of measuring instruments. Space and time were replaced with their definable variants: *distance/displacement* and *duration/interval*, which were quantified by the units of human-made instruments, such as rulers and clocks, thereby giving them practical meanings. As a result, the abstract concepts of space and time, defined as independent a priori from each other and from everything else (Kant 1998), have become constructed axioms of human thinking. The general appeal of space and time derives from the Newtonian framework of physics, referring to absolute time and absolute space, meaning that 'events' happen in a large 'theatre' (space) at some 'time' and any observer anywhere notices the same event at the same time. In our language-guided minds, time and space form the very basis of our imagination.

It would be hard to imagine and organize human life without clocks in the contemporary world. Time appears to be critical for all brain operations since every neuronal operation evolves 'in time', from coordination of muscle contractions during reaching to thinking and perhaps even dreaming (Tsao et al. 2022). Yet, when human-invented instruments are taken away, space and time revert back to dimensionless and ungrounded concepts. This happens because space and time, being immaterial, cannot exert an impact on things, including the body and the brain.

How are then brains of animals expected to ‘represent’ space and time and their relationship when they have no sensors for either space or time, and no clocks or rulers to inspect? Furthermore, neither clocks nor brains ‘make’ time, only imagined concepts of such a thing (Buzsáki & Llinás 2017). Finally, since space and time cannot be studied directly, they cannot be derived from first principles. Yet, as we will discuss below, contemporary neuroscience research is still performed and interpreted largely within the ‘theatre’ framework of classical physics, even though in contemporary physics space is defined as the things themselves and there is no time ‘in which’ events occur.

As an alternative, we suggest that organization of brain dynamics offers a constraint how the brain interprets the relationships among objects and events out in the world. The hippocampus, a postulated key player in spatial navigation and memory, is the largest neuronal ‘graph’ in the brain, which constantly generates neuronal sequences or trajectories (Pastalkova et al. 2008). We will discuss that time and space are emergent properties of higher-order sequence learning (Buzsáki and Tingley 2019; George et al. 2021). The hippocampus is ‘blind’ to sensory modalities, concepts of space and time and, instead, performs a singular algorithm that learns a sequential, relational structure of its ecological niche without a need for spatial or time inputs or resorting to Euclidean assumptions. Geometric concepts emerge from brain dynamics and therefore, space (and time) should not be the task (‘things-to-be-explained’) for neuroscience.

What Happened to Me, Where, and When?

For the topic of this volume, time and space are special because of their postulated roles in episodic memory. Cognitive neuroscience distinguishes two types of memories that we can verbally declared: memories for events, referred to as episodic (or autobiographic) memory, and memories for facts, referred to as semantic memory (Squire 1992; Tulving 1972). Episodic events have a duration and take place somewhere. In contrast to this time-directed and segment-defined type of memory, declarable semantic facts are abstracted punctate events; they define objects, living things, and facts in the surrounding world. Semantic memories are gradually formed from multiple overlapping episodes with common items (junctions) among the episodes (Buzsáki 2005; Buzsáki & Moser 2013; Nadel & Moscovitch 1997) and through the repetitions the ‘what’ becomes invariant to the temporal and spatial conditions of the individual episodes (Figure 3.1).

The separation and independence of the of *what*, *where*, and *when* and their recombination for recall appear to be an efficient economic solution, compared to an alternative in which every individual experience of our lifetime was stored separately in an extraordinarily long list (Eichenbaum 2004; Tulving 1972; Squire 1992). An episode is an unfolding storyline in Newtonian space-time coordinates.

For recall, we recombine the trajectory or sequence of events by multiplying the marginals, akin to the hypothesized process of generating colour by multiplying different portions of red, green, and blue wavelengths (Friston & Buzsáki 2016). By hypothesizing separate coordinates for the *what*, *where*, and *when*, it appeared that neuroscience has identified a road map for uncovering the neurophysiological mechanisms of episodic memory (Eichenbaum 2014). By designating hippocampal neurons as ‘place cells’ (O’Keefe and Nadel 1978), superficial entorhinal cortical neurons as ‘grid cells’ and calling the same neurons as ‘time cells’ (Eichenbaum 2014; Istkov et al. 2011) gave an impetus to this research programme. Yet, despite its appeal, this programme is based on an outdated framework and space-time conceptualization. To illustrate this caveat, it is useful to recapitulate briefly the evolution of thinking about space-time issues in physics and mathematics.

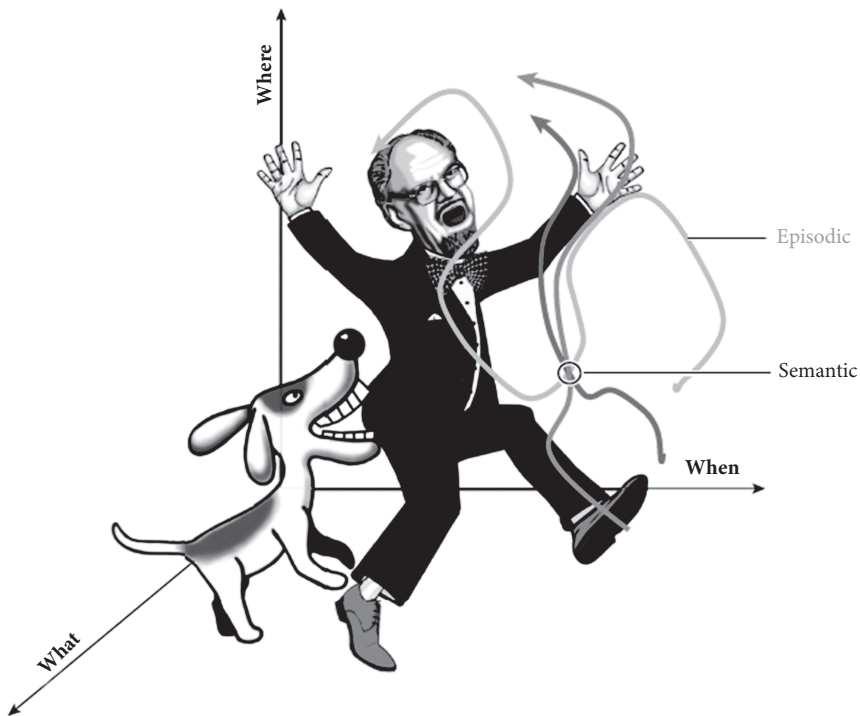


Figure 3.1 Space-time representation of episodic and semantic memories. Episodic memory evolves in space-time (what, where, when axes) with a corresponding neuronal trajectory (colour lines). The trajectory is a vector that unfolds in space-time (i.e., a segment with distance and duration). Only neuronal networks that generate such directional neuronal trajectories (as opposed to bidirectional correlations or associations) can support episodic memory. Three such trajectories (or episodes) are illustrated here. Intersection of multiple trajectories through the same space state can be considered as a node (black circle) or a snapshot in space-time (i.e., context-free semantic memory). Figure adapted from Buzsáki (2019).

Time of Physics and Time in the Brain

Modern science, including neuroscience, is quantitative: ‘measure what is measurable, and make measurable what is not so’ (https://www.goodreads.com/author/quotes/14,190.Galileo_Galilei). Measurements, in general, convert phenomena to numbers so that the observations can be written in the language of mathematics. In Newtonian physics, the objects on the scene of the theatre (i.e., space) interact instantly, assuming that the interaction speed is infinitely large. The Newtonian equations work reasonably well for describing the motion of the innermost planets of the solar system. However, if the interactions took time, that is if speed needs to be part of the model, the Newtonian model would no longer hold. It took two centuries to convincingly demonstrate that an observer can notice a distant event with a time delay instead of ‘at the same time.’¹ From this new vantage point, modern physics did not invalidate the classic science. Instead, it drew the limits of the approximation it used and derived an approximation with extended range of validity (i.e., it offered a more comprehensive theory) by pointing out that the interaction speed must be considered explicitly.

Hermann Minkowski, Einstein’s adviser in mathematics, wondered how the world would look like if the interaction speed between its objects were finite, although without specifying what the nature of the hypothetical limiting speed was (Pyenson 1977).² His model arose from experimental physics and through purely mathematical reasoning it led to modified ideas about space and time.³ The connection between physics and mathematics was the positive parameter c , which in the theory of special relativity was equated with the speed of light (Einstein 1905; Das 1993), emphasizing a finite speed of interaction. As in classical physics, where distance and duration are equated via speed and c is the key parameter in space-time in the relativity theory. In contrast, neuroscience still considers space and time as orthogonal dimensions, even though speed has been shown to be an important gain factor in neuronal computation (Buzsáki 2019). Space and time can refer to processes in the brain but also the animal’s actions. Speed in the brain has two components which relates to computing and transfer. Perhaps it is justifiable to entertain the idea that neuroscience can also exploit the Minkowski’s model by examining the critical role of c , such as conduction speed of axons and locomotion speed (as will be discussed below), at every level of neuroscientific inquiry, including the investigation of memory.

¹ Experienced anomalies, for example deviations from the ‘ideal’ laws of motion of celestial bodies, did not immediately lead to revisiting the model. Based on timing the eclipses of the Jovian moon Io, Römer suggested that the speed of light is finite (<https://www.amnh.org/learn-teach/curriculum-collections/cosmic-horizons-book/ole-roemers-speed-of-light>). But deviations rarely result in the rejection of a model. Only replacing the old with a more comprehensive model propels science forward.

² Minkowski’s model was a three and a half dimensional coordinate system, since the four coordinates were linked through the interaction speed (as an external parameter for the model; Minkowski 1908).

³ Euclidian geometry was no longer adequate to the task of describing physical reality (only an approximation), and had to be replaced by the geometry of a four-dimensional ‘world’: ‘Beyond the divisions of time and space which are imposed on our experience, there lies a higher reality, changeless, and independent of observer’ (Galison 1979).

Computing, Processing Neuronal Information, and Time Delays

The idea that the brain computes can be traced back to the computer metaphor of von Neumann (1993). As result, ‘computing’, ‘processing’, and ‘information’ have become frequently used words for describing observations in neuroscience experiments, although without a consensus what these terms actually refer to (Krakauer et al. 2017).

Von Neumann suggested that a computing element must have an ‘input section’ and an ‘output section’. In the case of chained (serial) operations, the data must be first transferred to the input section of one unit before performing an operation, and after the operation (‘transformation’ or ‘abstraction’ in neuroscience) the results are transferred from its output section to the input section of the next computing unit. All these changes evolve along a trajectory, where the rate of change can be conveniently compared to a reference clock time. In this formulation, the total operating time is separated into transfer time and processing (or computing) time (Mizuseki et al. 2011). In his classic computing paradigm, von Neumann omitted the transfer time but warned that with the changing technology (‘using much faster vacuum tubes’) the validity of that omission must be revisited. Further, he explicitly warned that applying his simplified paradigm ‘to neural computing would be unsound’ (*sic*; von Neumann 1993). However, he did not provide a solution for the general case of computing when the transfer time component not only cannot be omitted but even can dominate the total computing time, as is the case for the brain.

In the theory of special relativity all coordinates are distances and time is transformed to distance using the propagation speed of light, therefore the abstracted coordinate system is called ‘space-time’. In contrast, in modern electronics and in the brain, the primary feature is time (duration) and distances are measured along the path in which the signal is transformed to time, therefore, it can be called time-space system. Transfer in any kind of computing needs finite time because of the finite interaction speed in the implementation (Végő 2020, 2021; Végő and Berki 2022). The implication of time delays for time-space can be illustrated by a general computing process in the 4-D coordinate system, an extension of the famous ‘light propagation’ Gedankenexperiment (<http://visualrelativity.com/LIGHTCONE/minkowski.html>), supplemented with the need for a finite time to switch the light on (Figure 3.2). In the model, the light is emitted in the origin (the first computing unit) and the observer (the second computing unit) is located at the position marked by a circle. After the light is turned on, the second computing unit must wait until after ‘future light cone’ reaches observer’s position. The arrows at positions 1 and 2 show the timing relations for interaction speed differing by a factor of 2 and $\frac{1}{2}$, respectively.

The constraint (or perhaps evolutionary advantages) of temporal delays in neuronal computation is a poorly investigated territory of neuroscience. Just two hundred years ago, the vitalist thinking claimed that the speed of the mind is

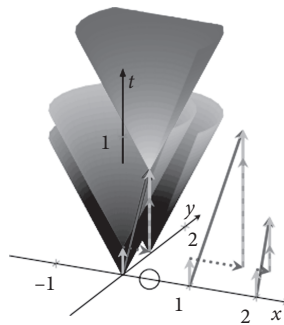


Figure 3.2 Critical importance of time delays in computation. The total computing time comprises components transfer and processing time. The processing occurs ‘in place’ (horizontal blue arrows) and the transfer between computing objects occurs ‘in time’ (vertical green arrows). The two operations mutually annihilate each other (mixed colour arrows), introducing inherent inefficiency into computing. The Minkowski-distance (red arrows) between the beginning and the end of computation differs for different signal propagation speeds. The computing time is a linear function of neither the transfer speed nor the processing speed. Reproduced from Végé (2021).

infinitely fast or, if it has a speed, it cannot be measured (*‘ignoramus et ignorabimus’*; Muller 1840), but soon von Helmholtz have demonstrated that the propagation of electricity in the nerve is in fact quite slow (von Helmholtz 1850). Subsequently it was established that the propagation speed of the action potential varies orders of magnitude in axons from tens of millimeters to meters/second, obeying a lognormal distribution (Buzsáki and Mizuseki 2014). An additional but small delay takes place in the chemical synapse (~ 0.3 ms conversion time from electrical to chemical signalling; Somjen 1972; Johnston and Wu 1994), leading to an analogue computation, measured as depolarization or hyperpolarization of the target membrane segment (Eccles 1957). A neuron integrates spike inputs from multiple upstream neurons in an analogue manner and reconverts them into a digital pulse code (action potentials or spikes) in the axon initial segment (Stuart et al. 1997). The transmission speed from the action potentials of upstream neurons to a downstream target neuron (*‘spike transmission delay’*) can vary substantially from a few milliseconds to hundreds of milliseconds, depending on the instantaneous conductance of the integrating neuron and the simultaneously acting feed-forward inhibition (Pouille and Scanziani 2001). Thus, while the axon conduction speed is constant, the efficacy of spike transmission varies across neurons and can vary as a function of the interaction of the neuron and the circuit in which the neuron is embedded. Therefore, the delay between Von Neumann’s ‘input section’ and ‘output section’ can vary, in contrast to digital computation. Such time-varying computation may be ‘most efficient in its use of resources is neither analogue computation nor digital computation but, rather, a mixture of the two forms’ (Strong et al. (1998). This distinction was already foreseen by von Neumann: ‘The language of the brain not the language of mathematics’ (von Neumann 2012).

The time delays in brain computation also impacts how the brain handles information. The mathematical framework introduced by Claude Shannon (1948) for

information transfer is connected to physical reality through an arbitrary external parameter, the ‘allowed signals of duration T ’ (aka, channel capacity). In Shannonian information theory interactions occur at the speed of light and in practice the mathematical handling corresponds to instantaneous interactions. Applying information theory to brain computation with its millions of times lower interaction speed and variable delays is therefore hard to justify (Shannon 1956; Végő and Berki 2022). Any communication, including electrical or neuronal, needs a carrier, such as an electrical impulse or neural spike. Because this carrier has a temporal component, the implication is that the information itself has a propagation speed. The speed of information propagation in the Shannon model is that of the electromagnetic wave and the information density is defined by the signal frequency. In contrast, in neural communication a spike alone only represents the beginning of a new message (as a synchronizing signal, it ‘wakes up’ the receiver), without any hint what ‘information’ is being transmitted. Single spikes are not able to deliver information, in line with the information theoretical conclusion that ‘if a source can produce only one particular message its entropy is zero’ (Shannon 1948). Instead, neurons use temporal intervals to communicate, and logical operations (information processing) in an analogue manner. Next, we will discuss how these considerations apply to neuronal computation.

Time Is Neuronal Space in the Brain

‘*Architecture makes music stand still. Here time turns into space*’ (Robert Kelly: ‘Phases of the Earth’; <https://www.visibleweather.com/home/tspace.html>). This poetic formulation of the relationship between space and time applies to the relationship between distance metric and duration metric in the brain. Simultaneity of two (or more) events may be deemed synchronous (i.e., occurring within a defined time interval of an observer) even if the two events occur at vastly different times. For example, action potentials arriving at the same time onto the dendrite of a reader neuron from a nearby and distant neuron exert a cooperative impact on discharge of the reader (target) neuron, even though the spikes in the two upstream neurons were generated tens of milliseconds apart (Figure 3.3A). Conversely, action potentials that are generated at the same (clock) time in a nearby and distant neuron, will arrive to the reader neuron tens of milliseconds apart (i.e., asynchronously; Figure 3.3B). This observer or ‘reader’-defined synchrony is critical in brain operations. If the action potentials from many upstream neurons arrive within the membrane time constant of the target (reader) neuron (τ : 10–50 ms for a typical pyramidal neuron), their combined action is cooperative because each of them contributes to the discharge of the reader neuron. Action potentials arriving later can only contribute only to initiating another action potential. Thus, from the reader neuron’s point of view, upstream partners that contribute to its spike discharge constitute a functional assembly (integrated by the membrane time constant), whereas spikes outside this time window can only be part of another assembly (Harris et al. 2003). This simple mechanism

can thus both integrate and segregate upstream neurons into discrete assemblies, irrespective of whether they are interconnected or not. The entire goal of synchronous cooperation is to have an impact, i.e., trigger an action potential in target neurons. Neurons which achieve this joint impact can be considered as a fundamental assembly. When the reader mechanism can integrate over longer time scales (e.g., NMDA channels or neuronal circuits), several fundamental assemblies can be concatenated.

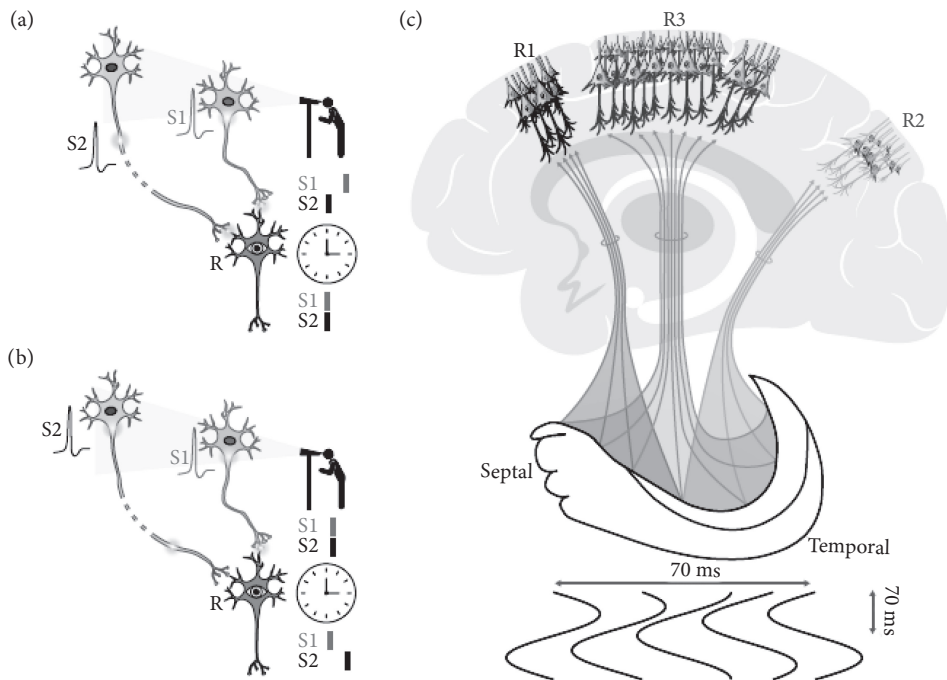


Figure 3.3 (A–C) Time is neuronal space. (A–B) Observer/reader-dependent definition of synchrony. (A) Spikes from upstream neurons (Sender 1 and Sender 2, S1, S2) arrive ‘at the same time’ to the dendrites of a reader neuron (R) but these spikes were initiated at different times by S1 and S2. (B) Spikes initiated at the same time in S1 and S2 arrive at different times to the Reader. The Reader-centric classification of spikes generalizes to populations as well. (C) Space-duration readout of neuronal messages. Spikes from the hippocampus that arrive within the time constant of the neocortical reader circuits R1 or R2 are integrated. Upstream (hippocampal) neurons which send these spike messages are considered separate assemblies by R1 and R2, respectively, due to timing of spikes and anatomical connectivity. The hypothetical reader circuit R3 has a ‘preformed anticipatory schema’ with a shared cipher (theta cycle frame) that helps to select features that are characteristic for anticipated events or activities and their change. Marking the message boundaries by cyclic inhibition partitions a continuous stream of events into discrete units and forms the hypothetical basis for extraction and synthesis of new information (a process called ‘abstraction’). In the example, the message sent from the entire hippocampus occurs in half a theta cycle (~70 ms). Reprinted from Buzsáki and Voroslakos (2023).

This reader-centric view becomes of utmost importance when one wants to understand communication across brain regions, such as the hippocampus and neocortex (Figure 3.3C). Because of the slow communication across neurons via the relatively slow conducting axons and charge time of the neuronal membrane, computation and messaging are not instant but protracted over time. This slow propagation of the activity within and across networks is reflected at the mesoscopic level as ‘traveling waves’ (Ermentrout and Kleinfeld 2001). For example, population activity travels from the dorsal to the ventral pole of the hippocampus in half a theta cycle (~70 ms; Figure 3.3C). Therefore, the question that arises when the entire hippocampus takes part in a particular computation is: how does the neocortical reader integrate (i.e., decode) neuronal messages from the hippocampus? As is the case in human language communication, where even the last word of the sentence can change the meaning of the sentence, the reader structure should know both the beginning and end of each message—in our example, the spikes from the entire hippocampal volume within 70 ms timeframes. Separation of networks into senders and receivers serves only didactic purposes. In the complex networks of the brain, most structures are bidirectionally connected and can serve as both senders and receivers, which functions can change rapidly by shifting the phases of the sender and receiving partners.

In summary, communication across multiple neuronal circuits occurs in a time-evolved manner. The term communication refers to an agreement between a sender and receiver. This ‘agreement’ is called the cipher known to both parties. A key aspect of the cipher is a method by which the messages are broken down into smaller information chunks or segments, the *modus operandi* by which the sent messages are transcribed and translated by the receiver. Both the sent and received messages must have a start and end time point and in these intervals a large neuronal ‘space’ (or volume) is involved. Chunking of messages by agreed rules allows the generation of and reading (i.e., ‘de-ciphering’) virtually infinite combinations from a finite number of elements in human, sign, body, artificial, and computer languages, music, and mathematical logic, and, presumably, in the brain. This small set of rules, usually referred to as syntax, governs the combination and temporal progression of discrete elements (such as letters or musical notes) into ordered and hierarchical relations (words, phrases, and sentences, or chords and chord progressions) that allow the generation of messages, which when interpreted by the receiver *becomes* information. Without a reader-interpreter neuronal ‘information’ does not have a meaning. Since messages are sent to multiple targets with varying distances and often distinct internal dynamics, the same exact action potential trains can be interpreted differently by the distinct readers. The potential substrate for a hypothetical neuronal syntax in the brain is the constellation of brain rhythms (Buzsáki 2010). The oscillation cycles serve as the cipher, marking the beginning and end of messages from short ‘neuronal letters’ (Harris et al. 2003), which are concatenated to become ‘neuronal words’ by slower rhythms by cross-frequency phase-amplitude coupling (Buzsáki and Draguhn 2004). The coordinating role of oscillation cycles

across brain regions assures that messages from multiple sources arrive at the 'same time' (phase), despite variable delays via multiple paths (Sirota et al. 2008).

Distance to Time Compression via Speed

In the rodent hippocampus, neuronal activity is organized by a 6–10 Hz theta oscillation (Buzsáki 2002) and its pyramidal neurons are believed to 'respond' to particular constellation of the environment ('place cells'; O'Keefe and Nadel 1978). The place fields of place cells are typically several times larger than the length of the rat, and place fields of several neurons overlap with each other over multiple theta cycles during ambulation. In addition to this locomotion-related overlap at the seconds scale, several place cells fire together in a given theta cycle such that the spike timing sequence of neuronal assemblies predicts the sequence of passed and upcoming locations in the rat's path, with larger time (phase) lags between spiking of place cells within the theta cycle representing larger distances (Figure 3.4A–C; Skaggs et al. 1996; Dragoi and Buzsáki 2006; Diba and Buzsáki 2008; Bouffard et al. 2018). In other words, if we take a 'snapshot' over a single theta cycle, the spike sequences (or 'neuronal trajectories', Figure 3.1) correspond to the trajectory of place fields the animal has just passed and is going to visit ('spatial segment' or distance; Figure 3.4). One interpretation of this relationship is that distances in the world are transformed to durations in the brain ('space-time' of physics) on the assumption that the phase of spiking of hippocampal neurons are 'driven' by some external cues. A problem with this interpretation is that the theta rhythm is induced in the brain and its phase varies independently from the external cues so that upon repeated runs the phase of theta varies randomly relative to the same spatial locations.

An alternative interpretation is that the primary mechanism is an internally maintained dynamics in the hippocampus, and this dynamics constrains how events in the world are matched to this pre-existing dynamics ('time-space'). In support of the latter interpretation, the time (theta phase) offsets in the hippocampus remain similar in different size environments, so that the same time segments (duration) between place cell spikes now correspond to larger distances in larger environments (Diba and Buzsáki 2008). From this perspective, the hippocampus can be viewed as a time-space zoom in which the neuronal resources are determined by an internal dynamics (the number of cell assemblies and the total number of spikes within theta cycles remain the same; Figure 3.4), therefore the resolution of the environment scales with its magnitude (in larger environments, the spatial resolution is poorer, with larger place fields and larger distances between place fields; Diba and Buzsáki 2008; Kjelstrup et al. 2008). What is the conversion mechanism in the brain between distance and duration?

In classical physics, distance and duration are equated via speed. One can expect that this general law also applies to organisms moving through space. As just discussed, each theta cycle contains a segment of travel (distance) coded as a sweep of

time (duration). How can this distance—duration relationship remain faithful when the animal runs at different velocities? If the rat traverses the place field of a neuron in 1 second during the first pass and then in half a second in the next pass, the place cell will be active for 8 and 4 theta cycles, respectively (assuming 8 Hz theta frequency). An important experimental observation is that the number of spikes within the place field remains the same even though the rat's velocity changes (Figure 3.4). For this reason, the number of spikes per theta wave and the overlap between spiking of successive place cells approximately doubles. Due to this speed-related firing rate gain per theta cycle, the magnitude of the cycle-to-cycle phase advance of spikes increases proportionally. Thus, speed gain compensates for the shorter time spent in the place field, leaving the relationship between theta phase and spatial position relatively invariant (Figure 3.4). Because the brain has constant access to velocity from the body and the vestibular system, time and distance travelled can be interchangeably calculated. Thus, the distance-to-duration compression may be only an appearance. Instead, the internally organized cell assembly sequences are available to match the animal's position to occasional landmarks but the occurrence of theta phase related spiking is not micromanaged by environmental cues but are controlled by internally generated dynamics.

The primacy of phase versus clock time (i.e., time related to a human-made device) is further supported by experiments in which the medial septum (a key structure that provides a 'pacemaker' theta signal to the hippocampus; Petsche et al. 1962) was cooled artificially to decrease the frequency of theta oscillation (Petersen and Buzsáki 2020). Cooling affected the distance-to-time, but not the distance-to-theta phase, conversion (Figure 3.4D, E). These findings revealed that cell assembly sequences in the hippocampus are not related to the time of reference clocks but to the phases of internally generated rhythms. In other words, brain 'time' and time measured by clocks, while they can be related, are independent.

Place Cells, Time Cells, Episode Cells, Tone Cells, and Many Other Names

A recent attempt to explain the role of time and relate it to space in cognitive phenomena is to distinct neuronal 'representations' of the future and the past (Tsao et al. 2022). According to this model, durations that start in the present and end in the future ('prospective timing') are supported 'clock-like mechanism that accurately tracks the continuing passage of time', either periodic, such as brain oscillations or cumulative. In contrast, durations that start in the past and end either in the past or the present ('retrospective timing'; 'episodic timing' or 'remembered duration') is 'computed only from memory of the interval and events occurring within it' (Tsao et al. 2022; Friedman 1993; Ornstein 1969). Thus, retrospective timing is 'the conversion of memory representations into a duration value' (Tsao et al. 2022). The main argument in favour of two distinct psychological times is that

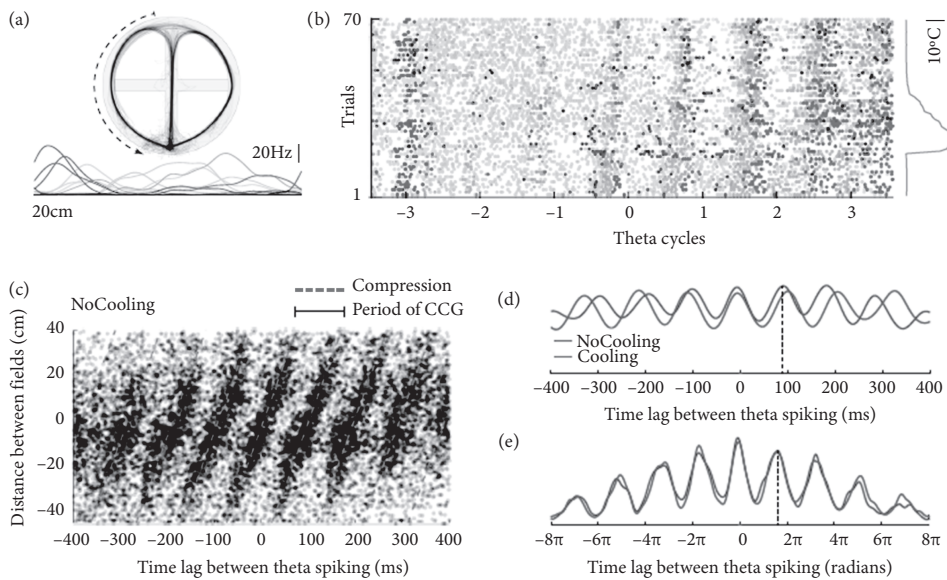


Figure 3.4 Distance-Theta Timescale Conversion. (A) Selected place fields in the left arm of the maze before medial septum cooling. Each coloured line is shows the smoothed firing rate of a place cell. (B) Within-theta cycle cell assemblies. Each row of dots is a trial of the spiking activity of place cells (same colour code as in (A) in successive theta cycles), organized into subsequent theta cycles, rather than distance or time. 0 is the theta cycle in the middle of the left arm. Note that cell assembly sequences within theta cycles (from dark blue to red) correspond to the order of place fields on the track. Note also that the phase preference of place cell spikes are preserved during medial septum cooling, (right trace), which manipulation decreased the oscillation frequency of the entire population. (C) Relationship between distances of place field peaks across neuron pairs (y axis) as the rat travels in the maze and their theta time-scale cross-correlogram lags (time lag on x axis) during control trials. Green dashed line; distance-time compression index (slope), $c = 0.54$ cm/ms. Note the reliable relationship between place field distances and time offset of spikes at theta (~100 ms) time scale. (D) Sum of the theta repeated spike cross-correlograms of place field pairs. The red line is the sum of all dots in (C) (NoCooling); the blue line is the sum of all dots of a similar graph during cooling. Vertical dashed line indicates that the oscillation frequency of the spike cross-correlation before cooling (~85 ms) is faster that the frequency of the population theta (~110 ms at 9 Hz). Note also that this relationship is increased during cooling. (E) Similar to (D), but instead of time lags, within-theta phase lags between spikes were calculated. Vertical dashed line indicates that the oscillation frequency of the spike cross-correlation is faster than the frequency of the population theta (2π). Note that the theta phase of spiking remained unaltered by slowing the theta frequency. After Petersen and Buzsáki (2020).

remembered time (retrospective) corresponds to ‘significantly different duration estimates for the same physical time intervals’ and the recall process is assisted by event boundaries that serve as time stamps or ‘nows’ (i.e., time points; Tsao et al. 2022). Unfortunately, this approach does not clarify what ‘tracking of elapsed’ time

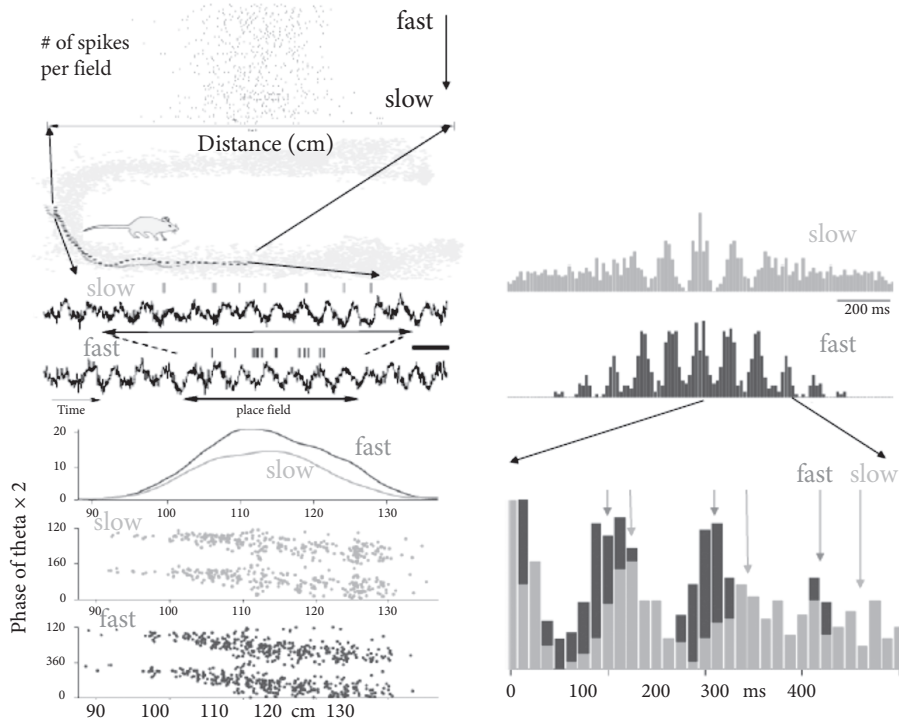


Figure 3.5 Speed-gain for distance-time compression in the hippocampus. (A) Top, The number of spikes (dots) within the neuron's place field is similar on slow and fast run trials. Trials are sorted by velocity from slowest to fastest. Middle, Movement trajectories of a rat through a place field on two trials with different speeds. Spikes (vertical ticks) and the corresponding theta rhythm of the same two trials (colour code). Horizontal arrows indicate the time it took for the rat to run through the place field. (B) Top, firing (place) field of the neuron during slow and fast runs (top half and bottom half speeds). Middle and bottom, Spike distributions within the place field as a function of the phase of the theta cycle during slow and fast runs. Note that the slope of the phase shift within the field remained the same, indicating that the position versus theta phase-spike relationship is preserved despite varying speed of the rat. (C) Oscillation frequency (shown here as spike autocorrelograms) of the same neuron during slow and fast trials. Note that the oscillation frequency of the place cell is faster during the faster trials. The duration $\times$ speed then is responsible for the preserved relationship between distance from the beginning of the place field and the rat's position. After Geisler et al. (2007).

by brain mechanisms actually mean. The tacit assumption is that through its sensors the brain can read genuine temporal clues and units that exist 'out in the world'. Unfortunately, there are no genuine *temporal* clues to read out there. Sunrise is a reliably recurring event that allows living things to anchor their biological processes but in itself is neither a 'time point' or duration. For retrospective timing, in contrast, the brain has to resort to reading its own dynamics without reference to outside events. Yet, it is hard to justify the assumption of distinct mechanism and a separation of

temporal phenomena. If the brain can generate its own time, why is there a need to postulate tracking of external time? Such attempts follow the dominant trends in current neuroscience that split complex problems arbitrarily, provide explanations to parts and attempt to generalize from the partial observations.

Another split of temporal phenomena is to distinguish time points (time ‘stamps,’ such as now, tomorrow, calendar dates, etc.) from duration (or interval) and use time points for temporal ordering of events (Fraisse 1963; Buhusi and Meck; 2005; Merchant et al. 2013). Notice that such practice also requires to refer to human-made instruments or to world events which themselves are not inherently temporal (such as spring or full moon). A deeper problem is the circular relationship between time as duration and time as an event stamp. Time point is the end or beginning of duration. Conversely, duration is arbitrarily defined by accumulation of units between two time points, analogous to the circular nature of distance and ‘here’ (i.e., a point is space; Buzsáki and Llinás 2017).

Sequential Brain Activity as a Foundation of Event Ordering

The above discussion of time can bring us back to navigation and to memory, both of which are inherently sequential, and thus may have a relationship to internally generated neuronal assembly sequences (Figure 3.1). There is a clear parallel spatial navigation and episodic memory, which can be referred to as mental or imagined navigation back into the past (Tulving 1972). A main difference is that mental navigation does not depend on immediate environmental or body-reafferent cues. It has been postulated that neuronal mechanisms, which initially depended on external cues in simple organisms, have become ‘internalized’ and disengaged from locomotion (Buzsáki 2005; Buzsáki and Moser 2013). Without external constraints, disengaged processing in complex brains can create an internalized virtual world and generate new knowledge through vicarious or imagined experience, tested against pre-existing and stored knowledge. This preservation of neuronal sequences can be demonstrated when a rodent is trained in a memory task without the translocation of the head and body. Figure 3.6 illustrates such an experiment in which rats learned to alternate between the left and right arms of a T maze. The key part of the experiment is that a delay is introduced between the choices during which the animal is required to run in a running wheel at a constant speed while facing the same direction on each run (Pastalkova et al. 2008). During running, theta oscillation is sustained and spiking activity of pyramidal neurons display a continuous sequence during the entire journey, with unique sequences for left and right choices. Thus, one can predict the animal’s arm choice in the maze tens of seconds before it acts out the plan with high accuracy, including maze arm choices on erroneous trials (Pastalkova et al. 2008). Thus, the unique sequences predict ‘what’ will happen. The duration, firing rate, theta phase offset relationships between neighboring neurons within the sequence are identical to those of place cells during ambulation in the maze arm.

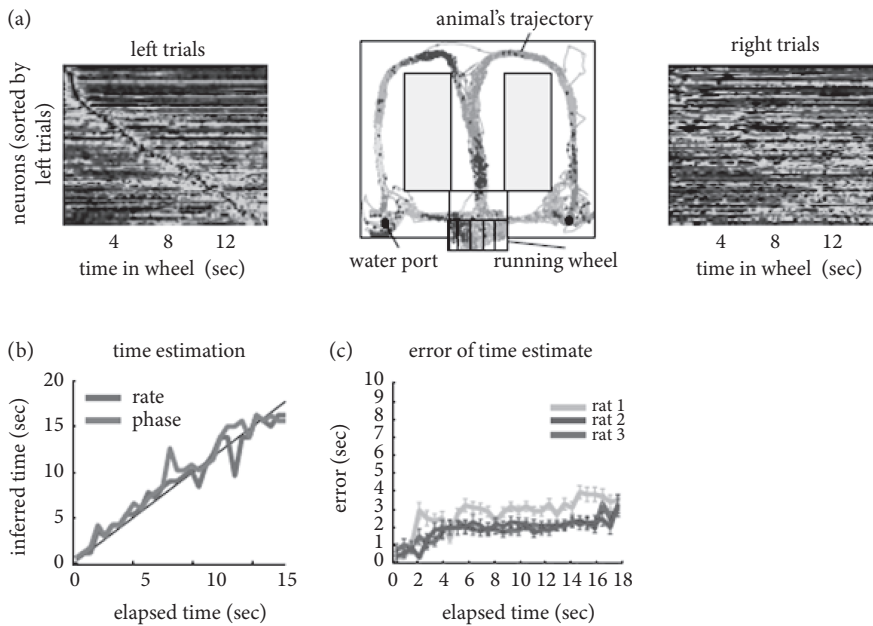


Figure 3.6 Time prediction from sequential neural activity in a memory task. (A) Spontaneous alternation task. The rat was required to make a correct choice for water reward on the basis of its previous choice (middle). Action potentials of different place cells are represented by coloured dots. Left and right, Average raster of firing rates for a population of hippocampal neurons during wheel running. Neurons are ordered according to the time of peak firing rate during left trials. Each line is a neuron, hot colours indicate increased firing rates. (B) Time detection on single trials using a time prediction model fit from all other trials. In each time bin, elapsed time in the running wheel is inferred either from the population firing rate vector (red) or the firing phases of active cells with respect to the theta oscillation (purple). In each case, the prediction approximates well the true time (black). (C) Error of time estimation from population vector of neuronal activity in three rats. Rat 1 had < 50 recorded neurons, rats 2 and 3 > 50 neurons. Note reliable estimation of running duration from neuronal activity. After Itskov et al. (2011).

From the evolving sequences, both in the wheel and the maze, one can also calculate the exact distance travelled by the animal and quantify the run distance against the units of a ruler. Alternatively, one can compare the neuronal sequences against the unit of a clock and report the elapsed time from the beginning of wheel running (Figure 3.6; Itskov et al. 2011). This relationship prompted Eichenbaum to declare that there are ‘time cells’ in the hippocampus (Eichenbaum 2014). For a moment, it appeared that the long search for the mechanism of episodic memory has been solved by identifying the *what*, *where*, and *when* in the hippocampus (Figure 3.1; Eichenbaum 2014). But this may be only an illusion. Note that in this prototype experiment, there is only one physical change, which is an internally generated sequence. When the same exact neuronal sequence or its individual members are compared with future arm turns, units of rulers and clocks, it appears that they ‘represent’ different things. But the three names refer to only one mechanism.

Here we need to refer back to the linear relationship between distance, duration and speed. As we have discussed above, knowledge about locomotion speed is critical for advancing neuronal assemblies at appropriate rate of change (Figure 3.5), and neurons, particularly inhibitory interneurons throughout the forebrain, vary their firing rates with speed (Czurko et al. 1999; Kropff et al. 2015). Distances can be computed from the product of the elapsed time and instantaneous speed. Vice versa, time can be computed from product of the distance travelled and instantaneous speed. Why do we expect then that brain circuits ‘code’ separately speed (‘speed cells’ Kropff et al. 2015), distance (O’Keefe and Nadel 1978), and time (Eichenbaum 2014), when any of these three variables can be computed from the remaining two? Coding for three separate variables with a linear relationship among them is redundant and would only lead to loss of precision.

Time Is Money: A Social Agreement

We started our overview with the postulate that brains neither sense or make immaterial space or time. Yet, throughout the discussion of the neuroscience observations we used these terms in their everyday meaning, as they refer to units of human-invented instruments. It would be hard to describe any experimental observations without resorting to space and time, the axioms of human thinking. But while physics have made strives to understand the content behind these terms, the vocabulary of neuroscience has remained in the framework of the Newtonian world.

If time is not perceived by the brain why do we feel that time pressure is our number one enemy? Perhaps no other human-invented artifact has had a larger impact on our lives than the clock. Over the past few millennia, time has become the most precious commodity. It all started by measuring distance and duration, that enabled to form a more concrete concept of space and time and their socially agreed metrics. In turn, the ensuing collective knowledge shaped the concept formation of individuals. While the lives of our hunter-gatherer predecessors were filled with timeless ghosts and deities, the invented instruments with their communicable units radically changed thinking of the emerging urban humans. With the help of created artifacts, humans can use their brains in a way that no other animal can approach and using language we can ‘fill’ the brains of our young with extraordinary amount of semantic knowledge, without the need to go through individual episodic experiences (Buzsáki et al. 2022).

To understand the apparent paradox between reality and feeling of reality, it may be instructive to relate time to another abstract yet more tangible extension of brain function: money. In a sense, money is an abstract representation of value. In its origin, money meant to facilitate exchange of goods when hunter gatherers settled and formed urban communities. Gold coins initially had the equivalent value of exchanged goods but, in today’s world, it is simply a symbolic or virtual value based on nothing else but a societal contract. Virtual money, such as a bank

account, is immaterial, and like space and time, cannot exert a direct effect on anything. It has no impact outside civilized humans. A hundred-dollar bill would mean nothing to humans who lived just a few hundred years ago and it has no value even today for isolated societies that live in the Amazon or the Australian desert, not to mention non-human species. Yet, the money concept is regarded as one of the strongest motivational forces, the primary driver of modern societies. But once the social contract for a particular form of currency ends, it becomes irrelevant. Similarly, if human-invented rods and clocks are removed from the arsenal of experimentation, space and time will revert back to dimensionless and ungrounded concepts.

Money has no meaning or utility for an isolated individual because it is not a product of a single brain. This may also be the case for space and time, which terms help communicate our experiences about world events to others. The concepts of space and time, likely money, are the product of a long-evolving societal agreement yet feel very real in our minds. Yet, I assume no neuroscientist would endeavour to study the impact of money per se on the brains of experimental animals. So why are we searching for space and time cells in their brains then? We may not find them. But we will find neuronal assembly sequences, which change at varying rates. Change can occur only to things, such as neurons, and things can exert an effect only on other things. Time is then an abstraction of change, not referring to anything concrete, similar to numbers that are immaterial, abstracted quantities. Instead for searching for mechanisms (*explanans*) of concepts (*explananda*) we created, we may try to revert our thinking by taking the brain as a starting point.

It may well be that self-organized brain networks just do what they can do, which is to generate sequentially active neuronal trajectories, which, in turn, can be matched to our executed (postdiction) and planned (prediction) actions rather than specifying time or space. These same internally generated sequences can index past experiences for recall in the right order or in an imagined or socially agreed coordinate system. The machinery that enabled navigation in the physical world have been exapted for more complex roles, including memory, planning, and imagination, yet navigation in real and mental space appear to use the same general purpose algorithm for tracking sequences of events and mapping the relationship between continuous variables (Buzsáki and Tingley 2018, 2023).

Actions of animals, simple or complex, are attempts to extract features of our niche and the resulting knowledge is constrained by species specific brain wiring and dynamics. Insects, birds, terrestrial and sea mammals may have very different world concepts, biased by the interactions between their brains and differing niches. Children below the age of 9 intuitively grasp the notion of speed but not time. 'Time relationships constructed by young children are so largely based on what they hear from adults and not on their own experiences' (Piaget 1946). Time is learned through language. The world views of people living across the Andes–Amazonia divide are quite different, as reflected by their distinct Quechua and Arawak languages, respectively (Green 2007). Living in open areas with visible and

identifiable mountain peaks facilitates landmark (allocentric) navigation, whereas if all one can see all the time is just trees in the jungle, egocentric navigation may be more efficient. The different actions taken by people in their own niche may inevitably shape language and communication style with others as well as concept formations about the world. Abstract space and time can afford concrete actions only when we measure them with our instruments using socially agreed units. Only in the context of these cultural innovations can one understand why space and time dominate our current thinking. However, these same innovations place researchers in a difficult situation when they are trying to find the ‘representations’ of space and time in the brain.

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